Supplementary Materials

Replication & Extension of Regional Favoritism (Hodler & Raschky, 2014)

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Abstract

This document reports the replication tables that accompany the main paper. The main paper covers Table II (main results), Table IV (geographic extent), and the 1992-2023 panel extension; the remaining five of the authors' seven main-text tables, Table I, Table III, Table V, Table VI, and Table VII, are reported here in full.

Keywords: Regional favoritism, Nighttime lights, Replication, Supplementary materials

1. Introduction

The main paper covers Table II (the main result), Table IV (the geographic-extent test), and the 1992-2023 panel extension. Hodler and Raschky (2014) (Hodler and Raschky, 2014) report seven main-text tables in total; the five this replication also reconstructed but the main paper does not cover are reported here instead: Table I's full descriptive statistics (the main paper reproduces only the two variables common to every specification), Table III's dynamics specification, Table V's determinants of the effect, Table VI's continent split, and Table VII's aid/oil heterogeneity test. Every specification below builds on the same fixed-effects regression described in the main paper's Section 4 (Empirical Framework), region and country-by-year fixed effects, standard errors clustered by leader spell, estimated on the authors' exact 126-country, 1992-2009 sample; that baseline is not restated per section. Data sources and construction deviations (G-Econ in place of Gennaioli et al., 2014; a reconstructed WVS-based FamilyTies measure; a more recent Penn World Table vintage for NationalGDP) follow the same conventions described in the main paper's Data section.

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2. Data

This section lists every data source behind the tables in this document and in the main paper, in full, since the main paper's Data section only has room to describe the sources behind Table II and Table IV. Sources are grouped as they enter the pipeline: the outcome variable, political-leader identification, geography, population, then the seven country-level covariates behind Table V, Table VI, and the extension's determinants and spillover tracks.

Table 1 covers every data source behind this document and the main paper, from the outcome variable through the seven country-level covariates. DMSP-OLS nighttime lights (Earth Observation Group, Payne Institute for Public Policy, Colorado School of Mines, 2013) is the dependent variable throughout, `log(light + 0.01)`, a small constant added because a large share of region-year observations sit at or near the lower bound. Archigos (Goemans et al., 2009) supplies the base universe of leader-spells (919 spells, 172 countries) but carries no birthplace of its own; PLAD (Bomprezzi et al., 2024) is the primary birthplace source, matching 86.9% of spells, and a Wikidata/Wikipedia scrape (Wikidata, 2026) fills a further 33 leaders where PLAD has zero coverage, bringing combined coverage to 90.5%. Geography and population are where this replication most directly diverges from the authors' own construction, discussed at length in the main paper's Data section (main paper Sections 3.3 and 3.4): the authors' CIESIN subnational boundaries (CIESIN, Columbia University, 2005) were never publicly released as underlying data, so this paper substitutes GADM 3.6 (GADM, 2018) throughout, and CIESIN's Gridded Population of the World, versions 3 and 4 (CIESIN, Columbia University and CIAT, 2005; CIESIN, Columbia University, 2018), is bridged into one annual series using their 2000 overlap year as a per-country scaling factor.

The seven country-level covariates used across Table V, Table VI, and the extension follow. Two are institutional: Polity's executive-constraints score (Marshall et al., 2010) and Barro-Lee's average schooling (Barro and Lee, 2013). Three are economic and cultural: the Penn World Table's GDP per capita (Heston et al., 2012), Alesina et al.'s linguistic fractionalization (Alesina et al., 2003), and Alesina and Giuliano's family-ties measure (Alesina and Giuliano, 2014). Two are resource-flow controls used in Table VII's aid/oil heterogeneity test: net official development assistance (OECD, 2013) and oil rents from the World Bank's adjusted net savings series (World Bank, 2013). The main paper's two documented construction deviations are both here: NationalGDP comes from a more recent Penn World Table vintage than the authors' own PWT 7.1, and FamilyTies is this project's own reconstruction from World Values Survey microdata (four of the authors' original six waves), since neither of Alesina and Giuliano's own papers publishes the underlying country lookup table.

None of these sources arrives pre-combined; each table above is the last link in a chain of merges the main paper's Data section does not have room to walk through. Birthplace identification runs Archigos and PLAD through two separate joins before a single leader-spell has a region: first Archigos's own leader-spell list is matched to PLAD by leader name and country-year to inherit a birthplace, then PLAD's coordinate (or the Wikidata gap-fill's, resolved the same way) is point-in-polygon joined to the GADM 3.6 ADM2 layer, so a spell only becomes usable once it has cleared both joins; the 90.5% combined coverage figure in Table 1 is the survival rate through that whole chain, not just the geocoding step. Building $Leader_{ict-1}$ itself is a further merge: a country-year's sitting leader (from the Archigos/PLAD chain) has to be matched against every region in that country in that year before the dummy can be defined for regions that are not the birth region, since a birth-region indicator with no comparison regions is not a panel variable. The population bridge merges GPWv3 and GPWv4 on their one shared year, 2000, computing each country's own GPWv4-to-GPWv3 ratio in that year and using it to rescale GPWv4's 2005 and 2010 values onto GPWv3's level before the two series are stacked into one 1990-2010 panel; a small number of countries where that ratio falls outside [0.85, 1.15], an implausible five-year jump, have GPWv4 left unscaled rather than forced onto a bridge factor likely driven by a source break rather than real growth. The seven Table V/VI covariates and the two Table VII covariates are simpler merges by comparison, a left join of each country-year covariate value onto every region-year row for that country in that year, which is also why Table V/VI/VII's "Number of regions" in Table 4, Table 5, and Table 6 is smaller than the full panel's 38,559: a region drops from a given column's regression only if that column's own covariate is missing for its country-year, so each column's own coverage gap, not a shared restriction, sets its region count.

Table 1: Data sources: outcome variable, political leaders, geography, population, and country-level covariates.

Source	Coverage	Used For
Outcome Variable		
DMSP-OLS Nighttime Lights	Annual 1992-2014, 0-63 DN, all countries; 30 arc-sec (~1km), WGS84, unprojected	Dependent variable throughout
Political Leaders		
Archigos, ext. w/ birthplaces	Leader per country-year to 2004, hand-ext. by HR to 2010; 919 spells, 172 countries	Defines who led which country and when; builds Experience/TotalTenure; no birthplace itself
PLAD Birthplace Coordinates	799/919 spells (86.9%); zero coverage for 25 countries	Primary birthplace source, point-in-polygon joined to GADM ADM2
Wikidata/Wikipedia Scrape	108 missing leaders queried, 42 resolved, 33 confirmed in-country	Gap-fill only where PLAD has zero coverage; combined coverage 832/919 = 90.5%
Geography		
Subnat. Boundaries → GADM 3.6	126 countries, 38,427 ADM2 regions; excludes countries under 500k pop. and territory above 65°N; vector polygons, WGS84, unprojected	Regional units of observation; authors' own CIESIN product never publicly released as data
Population		
Gridded Population v3 & v4	Region-year; GPWv3 anchors 1990/95, GPWv4 rev.11 anchors 2000/05/10, bridged and interpolated; GPWv3 2.5 arc-min, GPWv4 5 arc-min, WGS84, unprojected	Builds lnpop (log regional population) and per-capita light; 2000 overlap year sets the bridge factor
Institutions & Human Capital		
Political Regime Characteristics	Country-year, 1800-2010; rescaled Polity2 score, 0 to 1	Executive constraints and political competition; interacted with Leader in Table V
Educational Attainment Dataset	Country-year, 5-year intervals, 1950-2010	Average years of schooling; second institutional-quality proxy interacted with Leader
Economic & Cultural Covariates		
Penn World Table, GDP per Capita	Country-year, US dollars	Log GDP per capita, baseline economic control (this paper: a more recent PWT vintage than HR's own 7.1)
Linguistic Fractionalization	Country level, time-invariant	Probability two random citizens speak different languages; interacted with Leader
Family Ties	Country level, time-invariant	Strength of family bonds from World Values Survey; this paper reconstructs it from 4 of the authors' 6 waves
External Resource Flows		
Net Official Development Assistance	Country-year, current US dollars	Log ODA per capita; interacted with Leader in Table VII to test whether aid inflows fuel favoritism
Adjusted Net Savings, Oil Rents	Country-year, current US dollars per capita	Log oil rents per capita; interacted with Leader alongside aid in the same heterogeneity test

One further source, not in the table above because it substitutes for rather than supplements the authors' own data: Table II Column (8)'s regional GDP dependent variable. The authors use Gennaioli et al. (2014) (Gennaioli et al., 2014), a subnational regional GDP dataset that is not publicly released; this paper substitutes G-Econ 4.0 (Nordhaus et al., 2006), a 1-degree grid ($\sim$ 111km, WGS84, unprojected) gridded output dataset (190 countries, benchmark years 1990/95/2000/05) aggregated to ADM2 by point-in-polygon join. The two sources are not interchangeable in construction, so Column (8) is reported as indicative rather than a like-for-like replication, the same caveat carried through the main paper's Data and Results sections. Table 2 below reports the full observation, mean, and dispersion comparison between the authors' own panel and this paper's, for every variable in the table above.

3. S1. Table I: Descriptive Statistics (Full Sample)

Table 2 extends the main paper's two-variable descriptive comparison (Light$_{ict}$, Leader$_{ict-1}$) to the full nine-variable comparison from the authors' own Table I (p. 1007), including the five determinants used in Table V and the two spending controls, Aid$_{ct-1}$ and Oil$_{ct-1}$, in the authors' own layout: the standard deviation decomposed into overall, between (across regions or countries), and within (over time) components, following the standard panel "xtsum" convention, and, for the seven country-level determinants (Polity$_{ct-1}$ through Oil$_{ct-1}$), a second row in parentheses giving the same statistics collapsed to one observation per country-year rather than repeated once per region. The authors do not report a country-year row for Light$_{ict}$ or Leader$_{ict-1}$ (both are inherently region-year concepts), so neither does this table.

The two panels agree closely on Light$_{ict}$ and Leader$_{ict-1}$, as discussed in the main paper's Data section. Of the remaining seven variables, all seven means and standard deviations sit within a few percent of the authors' own, including NationalGDP$_{ct-1}$ and Oil$_{ct-1}$, despite NationalGDP coming from a different Penn World Table vintage than the authors used and no direct substitute being available for Oil's original construction; the level-of-development proxies these variables measure evidently do not depend heavily on which vintage or source constructs them. Observation counts are also close for six of the seven, but Aid$_{ct-1}$ is the exception: this paper's panel matches only 405,473 of the authors' 690,495 region-years, a 41% shortfall, by far the largest gap of any variable in the table, against 5-11% gaps everywhere else, including FamilyTies$_c$'s own incomplete reconstruction (covering only four of the six World Values Survey waves the authors' original source drew on; see the main paper's Data section for the full account of that limitation). Aid$_{ct-1}$ does not enter any specification reported in this document or the main paper, so this gap does not affect any reported result, but it is the largest single data-coverage difference between the two panels.

Table 2: Descriptive statistics, HR 2014's own layout (p. 1007): region-year panel, 126 countries, 1992-2009. SD is overall, between, within (xtsum decomposition). Rows in parentheses are the same statistics at the country-year level, for Polity$_{ct-1}$ through Oil$_{ct-1}$ only.

	Authors (126 countries, 1992-2009)					This Paper (same 126 countries, 1992-2009)				
Variable	Obs.	Mean	SD (overall, between, within)	Min	Max	Obs.	Mean	SD (overall, between, within)	Min	Max
Light$_{ict}$	690,495	0.050	2.482, 2.425, 0.538	-4.605	4.143	655,234	0.097	2.456, 2.398, 2.497	-4.605	4.143
Leader$_{ict-1}$	690,495	0.004	0.060, 0.045, 0.040	0.000	1.000	654,254	0.004	0.060, 0.046, 0.040	0.000	1.000
Polity$_{ct-1}$	684,213	0.789	0.257, 0.239, 0.099	0.000	1.000	612,321	0.789	0.258, 0.259, 0.090	0.000	1.000
(country-year)	(2,205)	(0.670)	(0.312, 0.288, 0.123)			(1,917)	(0.679)	(0.307, 0.293, 0.108)		
Schooling$_{ct-1}$	648,240	8.168	2.783, 2.724, 0.570	0.925	13.022	586,029	8.124	2.778, 2.740, 0.568	0.930	13.052
(country-year)	(1,922)	(6.968)	(2.920, 2.891, 0.518)			(1,700)	(7.081)	(3.015, 2.973, 0.519)		
NationalGDP$_{ct-1}$	683,669	8.868	1.147, 1.137, 0.151	5.080	10.855	615,056	8.829	1.178, 1.156, 0.147	5.081	10.855
(country-year)	(2,200)	(8.212)	(1.347, 1.337, 0.180)			(1,941)	(8.243)	(1.358, 1.338, 0.166)		
Language$_c$	679,119	0.318	0.281, 0.281, 0.000	0.002	0.923	647,397	0.312	0.286, 0.284, 0.000	0.002	0.923
(country-year)	(2,161)	(0.438)	(0.298, 0.298, 0.000)			(2,017)	(0.441)	(0.297, 0.297, 0.000)		
FamilyTies$_c$	551,004	0.096	0.311, 0.311, 0.000	-0.851	0.606	521,375	0.059	0.365, 0.361, 0.000	-0.912	0.572
(country-year)	(1,112)	(0.041)	(0.368, 0.371, 0.000)			(1,016)	(-0.017)	(0.391, 0.392, 0.000)		
Aid$_{ct-1}$	690,495	6.191	4.743, 4.069, 2.440	-10.632	13.712	405,473	7.887	3.284, 2.087, 2.649	-10.597	13.559
(country-year)	(2,251)	(7.794)	(4.746, 4.348, 1.931)			(1,471)	(9.256)	(2.653, 2.044, 1.983)		
Oil$_{ct-1}$	645,396	9.357	4.116, 4.457, 1.001	0.000	16.396	609,481	8.950	4.035, 3.888, 1.184	0.000	16.126
(country-year)	(1,820)	(6.590)	(5.262, 5.312, 1.191)			(1,895)	(5.964)	(5.147, 4.925, 1.478)		

4. S2. Table III: Dynamics of Regional Favoritism

Table 3 reports the authors' event-study-style specification (p. 1014-1015), testing whether the birth-region effect is genuinely tied to a leader's tenure rather than a pre-existing regional trend, in HR 2014's own row order and variable names. Column (1) adds a one-year placebo window (Future1$_{ict-1}$, Past1$_{ict-1}$: was this region about to become, or did it just stop being, the leader's birth region). Column (2) replaces that with a three-year window (Future3$_{ict-1}$, Past3$_{ict-1}$, plus the linear trends Pretrend$_{ict-1}$ and Posttrend$_{ict-1}$ over that window); these two columns test for a pre-existing or lingering regional trend, not a fixed-effects specification choice, so neither is a plain baseline the later columns build on. Columns (3)-(5) keep the three-year placebo window and add interactions of the birth-region dummy with Experience$_{ct-1}$ (years the leader has held office so far) and $\text{TotalTenure}_{ct-1}$ (the spell's eventual total length), testing whether the effect grows with time in office.

An earlier version of this table (and the script generating it) had guessed this column layout from term counts alone, rather than reading it directly off HR 2014's own Table III: it put Leader_{t-1} alone in Column (1) and all four placebo dummies together in Column (2). Checked directly against the authors' own PDF (p. 1015), that guess was wrong on both columns, and the sign on Leader × TotalTenure in Column (5) was transcribed as positive when the authors report it as negative (-0.002). Both are corrected here, and the model is re-estimated on the corrected specification.

The replication's placebo terms stay statistically insignificant in both Columns (1) and (2), matching the authors' own finding that there is no detectable pre-trend. The baseline Leader_{t-1} coefficient stays close to the authors' corresponding value in every column (all five of the authors' own Leader_{t-1} estimates, 0.038 down to 0.003 across Columns 1-5, are matched within one standard error by this replication's own; Column (1)'s own estimate, 0.038, now matches the authors' to three decimals under the corrected specification).

The tenure-accumulation channel is where the two panels part ways. The authors report a positive, significant Leader × Experience effect in both Columns (3) and (5) (0.007 and 0.009, both $p<0.01$), and a positive, significant Leader × TotalTenure effect in Column (4) (0.005, $p<0.01$), but their own Column (5) TotalTenure interaction is small, negative, and not itself significant (-0.002, SE 0.003), so the authors' own evidence for a TotalTenure channel rests on Column (4) alone, not on both columns where it appears. This replication's corresponding interactions are smaller in every column and not statistically significant anywhere, though they carry the same sign as the authors' Column (4) throughout. The divergence is specific to this tenure-accumulation channel, not to the core favoritism effect.

Reading the columns economically. Columns (1)-(2) are an identification check, not a substantive result in their own right: a region that lights up before its leader takes office, or stays bright after he leaves, would say the effect is driven by something already underway in that region rather than by the leader's own presence in power, undermining the causal story. Both placebo windows coming back at essentially zero, in both panels, is the paper's evidence that favoritism is switched on and off with tenure itself, not a proxy for some slower-moving regional trend. Columns (3)-(5) then ask an economic question about the mechanism: does favoritism behave like a stock a leader builds up, or a flow that starts immediately and stays flat? Experience is the leader's own political capital accumulated so far, a measure of how consolidated his hold on power is at any given point in his term; TotalTenure is that same spell's eventual length, known only in hindsight, so it proxies less for consolidation in real time than for whatever let this particular leader stay in office as long as he did (strong patronage networks, weak term limits, favorable political conditions). HR's own positive, significant coefficients on both suggest favoritism grows the longer and more securely a leader holds office, consistent with a patronage relationship that deepens with tenure rather than one struck once at the start; this replication's much weaker, insignificant version of the same interactions does not contradict that story so much as fail to detect it with the same confidence, given the same-sign point estimates throughout.

Table 3: Table III replication, HR 2014's dynamics specification, in the main paper's Orig./Repl. layout and the authors' own row order (p. 1015). Dependent variable: Light_{ict} throughout. Standard errors in parentheses, clustered by leader spell.

	(1)		(2)		(3)		(4)		(5)	
	Light_{ict}		Light_{ict}		Light_{ict}		Light_{ict}		Light_{ict}	
	Orig.	Repl.	Orig.	Repl.	Orig.	Repl.	Orig.	Repl.	Orig.	Repl.
Leader_{t-1}	0.038**	0.038**	0.039*	0.038**	0.006	0.012	0.003	-0.005	0.017	0.005
	(0.016)	(0.016)	(0.021)	(0.015)	(0.024)	(0.021)	(0.028)	(0.022)	(0.028)	(0.025)
Leader $\times$ Experience_{ct-1}					0.007***	0.005**			0.009***	0.004
					(0.002)	(0.002)			(0.003)	(0.005)
Leader $\times$ $\text{TotalTenure}_{ct-1}$							0.005***	0.005***	-0.002	0.002
							(0.002)	(0.002)	(0.003)	(0.005)
Future1_{ict-1}	-0.005	0.004								
	(0.029)	(0.023)								
Future3_{ict-1}			0.002	0.006	0.006	0.001	0.007	0.002	0.006	0.001
			(0.025)	(0.016)	(0.031)	(0.035)	(0.031)	(0.035)	(0.031)	(0.035)
Pretrend_{ict-1}					-0.005	-0.001	-0.005	-0.001	-0.005	-0.001
					(0.018)	(0.014)	(0.018)	(0.014)	(0.018)	(0.014)
Past1_{ict-1}	0.011	-0.005								
	(0.024)	(0.020)								
Past3_{ict-1}			0.002	0.013	0.007	-0.032	0.007	-0.015	0.006	-0.028
			(0.020)	(0.014)	(0.027)	(0.028)	(0.027)	(0.028)	(0.027)	(0.030)
Posttrend_{ict-1}					-0.001	0.021	-0.001	0.014	-0.001	0.019
					(0.012)	(0.014)	(0.012)	(0.015)	(0.012)	(0.015)
Number of regions	38,427	38,559	38,427	38,559	38,427	38,474	38,427	38,474	38,427	38,474
Observations	690,495	613,433	690,495	613,433	690,495	575,575	690,495	575,575	690,495	575,575
R^2 / Within R^2	0.319	0.475	0.319	0.475	0.319	0.359	0.319	0.359	0.319	0.359
Region FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country-Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes. Fixed effect regressions using annual data for subnational regions between 1992 and 2009. Light_{ict} is the log of average nighttime light intensity plus 0.01. Leader_{ict} is a dummy variable equal to 1 if region i is the birth region of the political leader in country c in year t, and 0 otherwise. Experience_{ct} is the number of years the political leader has been in power until year t. TotalTenure_{ct} is the total number of years the political leader was in power up to 2010. Future1_{ict} is a dummy variable equal to 1 if region i is the birth region of the political leader in $t+1$, but not in t; and 0 otherwise. Future3_{ict} is a dummy variable equal to 1 if region i is the birth region of the political leader in $t+1$, $t+2$, or $t+3$, but not in t; and 0 otherwise. Pretrend_{ict} is a time trend for the years in which $\text{Future3}_{ict} = 1$. Past1_{ict} is a dummy variable equal to 1 if region i is the birth region of the political leader in $t-1$, but not in t; and 0 otherwise. Past3_{ict} is a dummy variable equal to 1 if region i is the birth region of the political leader in $t-1$, $t-2$, or $t-3$, but not in t; and 0 otherwise. Posttrend_{ict} is a time trend for the years in which $\text{Past3}_{ict} = 1$. Standard errors are adjusted for leader clustering. ***, **, * indicate significance at the 1%, 5%, and 10% levels, respectively. (HR 2014, Table III note, p. 1015.) This replication's own additions: Columns (1)-(5) all use this project's own boundary-lag-corrected Leader_{ict-1} (Section 3.5 of the main paper) and its own construction of Experience/TotalTenure from PLAD/Archigos spells, not the authors' own underlying data. Column (1) is Leader_{t-1} + Future1 + Past1; (2) is Leader_{t-1} + Future3 + Past3 + Pretrend + Posttrend; (3)-(5) add the Experience and/or TotalTenure interactions on top of Column (2)'s specification. Orig. R^2 above is the authors' own reported R^2; Repl. reports this project's within-R^2 on the HR/Stata convention (a region-FE-only null model), which is not directly comparable to the authors' figure in level.

5. S3. Table V: Determinants of Regional Favoritism

Table 4 reports the authors' cross-sectional determinants specification (p. 1020-1022), interacting the birth-region dummy with five country characteristics one at a time (Columns 1-5) and jointly (Column 6): Polity (a democracy index), Schooling, NationalGDP, Language (linguistic fractionalization), and FamilyTies. A negative interaction means the favoritism effect is smaller in countries scoring higher on that characteristic.

The replication's Column (1) reproduces the authors' central finding most clearly: the Leader $\times$ Polity interaction is negative and statistically significant in both panels, of a similar magnitude, consistent with favoritism being weaker in more democratic countries. The Leader $\times$ Schooling interaction (Column 2) keeps the authors' sign, both panels finding weaker favoritism where average schooling is higher, but loses significance in the replication (the authors' own estimate is significant at the 1% level, this paper's at neither level). NationalGDP's interaction (Column 3) also keeps the authors' sign but loses significance the same way. Language's interaction (Column 4) is the closest match of the five, both sign and significance holding up. FamilyTies (Column 5) is the one that reverses: the authors report a positive, significant interaction, while this replication's is indistinguishable from zero. These gaps follow from the same source-construction issues noted in Section 3 (NationalGDP from a different PWT vintage, FamilyTies from an incomplete WVS reconstruction), though the reversal on FamilyTies specifically is larger than the Table I comparison alone would predict.

Column (6)'s combined specification keeps the Polity interaction as the most stable finding across both panels; the other four covariates lose significance in the replication once they compete with Polity for the same underlying cross-country variation, a pattern the authors' own Column (6) shows to a lesser degree. The one term that changes qualitatively in Column (6) is the baseline Leader_{t-1} coefficient itself: the authors' own Column (6) estimate is small, negative, and not significant (-0.036, SE 0.134), while this replication's is larger, also negative, and marginally significant (-0.295, SE 0.175), the same sign, but only the replication's crosses conventional significance. Because five interaction terms and their lower-order components enter the same specification, this sign reversal (relative to Columns 1-5, where Leader_{t-1} stays positive in both panels) is best read as a symptom of collinearity among the five determinants rather than a substantive finding on its own.

An earlier version of this table had transcribed the authors' own Column (6) Leader_{t-1} value as positive (0.036); checked directly against HR 2014's own Table V (p. 1021), it is negative. This is corrected below, along with the row order (the authors' own table lists one row per determinant, Leader × Polity through Leader × FamilyTies, each populated in its own column and again in Column (6), rather than a single generic "Leader × X" row) and the full per-column Number of regions, Observations, and R^2 footer, which the authors report varies column to column (684,213 down to 520,081) rather than staying fixed.

Reading the columns economically. Each column asks whether a different feature of a country's political economy amplifies or damps the birth-region effect. Column (1)'s Polity interaction is the paper's institutional-constraints story: in a country with strong checks and balances, a leader who tried to funnel resources toward his own region would face opposition parties, an independent judiciary, or a free press able to expose and contest it, so the favoritism this paper measures should be smaller precisely where those constraints bind, which is what the negative interaction says in both panels. Column (2)'s Schooling interaction reads similarly but through a different channel, a more educated population being better able to monitor and punish patronage at the ballot box or through civil society, so more schooling predicting less favoritism is the same accountability story from the citizen side rather than the constitutional side. Column (3)'s NationalGDP interaction is more ambiguous by construction: a richer country could show less favoritism because it needs less (public goods are already broadly provided) or because it has stronger institutions that happen to correlate with income, or it could show more because a bigger fiscal pie gives a leader more to redirect. Column (4)'s Language interaction is a coalition-building story: in a linguistically fractionalized country, a leader plausibly relies more on visible, targeted transfers to hold together a coalition across groups, so favoritism being stronger where fractionalization is higher is consistent with regional patronage substituting for the national cohesion a shared language would otherwise provide. Column (5)'s FamilyTies interaction is a particularism story in the same spirit: where kinship and family obligation carry more weight than universalistic norms, a leader's own extended network is a more natural conduit for resources, so stronger family ties predicting more favoritism fits a clientelist mechanism operating through personal rather than institutional channels. Column (6) asks which of these five channels survives once they all compete for the same explanatory power, and in both panels it is the institutional one, Polity, that does, consistent with formal political constraints being a more fundamental brake on favoritism than any of the four cultural or economic proxies.

Table 4: Table V replication, HR 2014's determinants specification, in the main paper's Orig./Repl. layout and the authors' own row order (p. 1021). Standard errors in parentheses, clustered by leader spell.

	(1)		(2)		(3)		(4)		(5)		(6)	
	Light_{ict}		Light_{ict}		Light_{ict}		Light_{ict}		Light_{ict}		Light_{ict}	
	Orig.	Repl.	Orig.	Repl.	Orig.	Repl.	Orig.	Repl.	Orig.	Repl.	Orig.	Repl.
Leader_{t-1}	0.262***	0.170**	0.119***	0.088**	0.196**	0.134	-0.008	-0.011	0.008	-0.007	-0.036	-0.295*
	(0.056)	(0.066)	(0.040)	(0.043)	(0.082)	(0.092)	(0.017)	(0.016)	(0.012)	(0.014)	(0.134)	(0.175)
Leader × Polity_{ct-1}	-0.298***	-0.170**									-0.240***	-0.050
	(0.063)	(0.072)									(0.065)	(0.086)
Leader × Schooling_{ct-1}			-0.012***	-0.007							-0.024***	-0.009
			(0.004)	(0.005)							(0.007)	(0.007)
Leader × $\text{NationalGDP}_{ct-1}$					-0.019**	-0.011					0.050***	0.045**
					(0.009)	(0.010)					(0.018)	(0.019)
Leader × Language_{c}							0.120***	0.123**			0.016	0.053
							(0.040)	(0.050)			(0.052)	(0.048)
Leader × FamilyTies_{c}									0.063**	-0.000	0.035	0.050
									(0.032)	(0.032)	(0.034)	(0.034)
Number of regions	38,427	38,380	36,033	36,316	38,179	38,310	37,795	38,061	30,631	30,284	29,123	28,792
Observations	684,213	608,062	648,240	581,963	683,669	610,749	679,119	606,255	551,004	490,880	520,081	467,301
R^2	0.320	0.975	0.330	0.975	0.320	0.975	0.318	0.976	0.308	0.972	0.313	0.974
Region FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country-Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes. Fixed effect regressions using annual data for subnational regions between 1992 and 2009. Leader_{ict-1} is a dummy variable equal to 1 if region i is the birth region of the political leader in country c in year $t-1$, and 0 otherwise. Polity_{ct-1} is the Polity2 score, rescaled 0 to 1. Schooling_{ct-1} is average years of schooling. $\text{NationalGDP}_{ct-1}$ is the log of GDP per capita. Language_c is an index of linguistic fractionalization. FamilyTies_c is a measure of the strength of family ties. Standard errors are adjusted for leader clustering. ***, **, * indicate significance at the 1%, 5%, and 10% levels, respectively. (HR 2014, Table V note, p. 1021, adapted.) This replication's own additions: Column (6) interacts the birth-region dummy with all five determinants jointly, so each determinant's row is populated in its own single column and again in Column (6), not a generic "Leader x X" row. NationalGDP and FamilyTies are built from different underlying sources than the authors' own (Section 2). Orig. R^2 is the authors' own reported R^2; Repl. reports fixest's overall (not within-transformed) R^2, which nets out the region and country-year fixed effects and is mechanically much closer to 1, so the two are not directly comparable in level.

6. S4. Table VI: Favoritism by Continent

Table 5 decomposes the pooled Leader_{t-1} effect into five continent-specific terms (p. 1023): Africa, Americas, Asia, Europe, and Oceania, with no omitted reference category since region and country-year fixed effects already absorb any continent-level main effect. Column (1) is the plain decomposition; Column (2) adds the Polity interaction; Column (3) adds all five Table V determinants.

Column (1) does not reproduce the authors' geographic pattern as closely as Columns (2)-(3) reproduce their covariate pattern. Asia, the authors' largest and most significant continent effect (0.121, p<0.01), loses its significance entirely in this replication (0.036, not significant), the largest single change in the table. Africa stays significant in both panels but at a lower confidence level here (authors' 0.071, p<0.01, against this paper's 0.089, p<0.05). Europe's sign matches, small and negative in both the authors' data (-0.019, p<0.1) and here (-0.017, p<0.1), both only weakly significant; an earlier version of this table had transcribed the authors' Column (1) Europe and Oceania values as positive, which, checked against HR 2014's own Table VI (p. 1023), is wrong for both; Oceania is -0.112 (not significant) in the authors' own data as well, also matching this replication's sign. Americas and Oceania are not significant in either panel. Once covariates enter in Columns (2)-(3), a second, systematic divergence appears: every one of the authors' five continent coefficients in Column (3) is positive, while every one of this paper's is negative, the same full sign reversal Column (6) of Table V shows for the pooled Leader_{t-1} term once all five determinants enter together (Section 5). Column (2), with only the Polity interaction added, does not show this reversal, so it is specific to the fully saturated Column (3) specification, consistent with collinearity among the five determinants rather than five independent continent-level findings. The Polity interaction itself stays the most stable term across both panels throughout Columns (2)-(3), negative and significant in every case though smaller in magnitude here. Among Column (3)‘s other four determinants, the authors’ own NationalGDP interaction is positive (0.047**) rather than negative; an earlier version of this table’s footnote had it as negative, corrected below.

Reading the columns economically. Column (1)‘s five continent coefficients are a first, unconditional look at where in the world favoritism runs strongest, before any attempt to explain why. Asia and Africa carrying the largest, most significant effects in the authors’ own data (and Africa still significant, if smaller, in this replication) is consistent with the countries in HR’s sample that fall in those continents skewing toward weaker political institutions and more discretionary, personalist rule, exactly the conditions Column (1) of Table V already flags as predicting more favoritism; Europe’s small, weakly significant coefficient (whichever sign either panel finds) fits the same story from the other side, since HR’s European sample leans toward the stronger-institution countries Table V’s Polity interaction associates with less favoritism. Column (2) tests whether that geographic pattern survives once Polity is held constant within each continent, and it does not: every continent’s coefficient jumps to a similar, large, significant magnitude once Polity enters, which says the geographic pattern in Column (1) is not five independent regional cultures of favoritism so much as one institutional-quality gradient that happens to correlate with continent, the raw continent effects in Column (1) proxying for Polity rather than adding anything Polity does not already capture. Column (3) pushes this further by adding all five Table V determinants at once, and the continent coefficients lose their earlier sign and significance entirely, the same collinearity pattern Table V’s own Column (6) shows for the pooled Leader$_{t-1}$ term: once schooling, income, language, and family ties compete with Polity for the same underlying cross-country variation, no single term, continent included, retains a clean, independent read on what actually drives favoritism.

Table 5: Table VI replication, HR 2014's continent-split specification, in the main paper's Orig./Repl. layout and the authors' own row order (p. 1023). Standard errors in parentheses, clustered by leader spell.

	(1)		(2)		(3)	
	Light_{ict}		Light_{ict}		Light_{ict}	
	Orig.	Repl.	Orig.	Repl.	Orig.	Repl.
Leader × Africa_c	0.071***	0.089**	0.235***	0.180***	0.041	-0.261
	(0.026)	(0.034)	(0.047)	(0.064)	(0.167)	(0.163)
Leader × Americas_c	0.000	0.051	0.243***	0.183**	0.056	-0.151
	(0.025)	(0.032)	(0.067)	(0.082)	(0.179)	(0.187)
Leader × Asia_c	0.121***	0.036	0.296***	0.139*	0.005	-0.238
	(0.042)	(0.036)	(0.073)	(0.082)	(0.147)	(0.176)
Leader × Europe_c	-0.019*	-0.017*	0.239***	0.127	0.035	-0.215
	(0.010)	(0.009)	(0.067)	(0.079)	(0.163)	(0.178)
Leader × Oceania_c	-0.112	0.014	0.106	0.141*	0.167	-0.068
	(0.077)	(0.046)	(0.101)	(0.085)	(0.168)	(0.182)
Leader × Polity_{ct-1}			-0.278***	-0.151*	-0.252***	-0.073
			(0.070)	(0.081)	(0.068)	(0.086)
Leader × Schooling_{ct-1}					-0.027***	-0.013*
					(0.007)	(0.007)
Leader × $\text{NationalGDP}_{ct-1}$					0.047**	0.039*
					(0.020)	(0.020)
Leader × Language_c					0.024	0.096*
					(0.046)	(0.049)
Leader × FamilyTies_c					0.011	0.007
					(0.037)	(0.039)
Number of regions	38,427	38,559	38,427	38,380	29,123	28,792
Observations	690,495	613,433	684,213	608,062	520,081	467,301
R^2	0.319	0.975	0.320	0.975	0.313	0.974
Region FE	Yes	Yes	Yes	Yes	Yes	Yes
Country-Year FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes. Fixed effect regressions using annual data for subnational regions between 1992 and 2009. Leader_{ict-1} is a dummy variable equal to 1 if region i is the birth region of the political leader in country c in year $t-1$, and 0 otherwise. Africa_c, Americas_c, Asia_c, Europe_c, and Oceania_c are dummy variables for the respective continents. Polity_{ct-1} is the Polity2 score, rescaled so that it ranges from 0 to 1, with higher values implying stronger political institutions. Schooling_{ct-1} is the average years of schooling attained. $\text{NationalGDP}_{ct-1}$ is the log of GDP per capita. Language_c is the index of linguistic fractionalization. FamilyTies_c is a measure of the strength of family ties. Standard errors are adjusted for leader clustering. ***, **, * indicate significance at the 1%, 5%, and 10% levels, respectively. (HR 2014, Table VI note, p. 1023, adapted.) This replication's own additions: no continent is an omitted reference category, since region and country-year fixed effects already absorb any continent-level main effect; each continent's coefficient is a level effect in its own right, not a contrast against a baseline continent. Orig. R^2 is the authors' own reported R^2; Repl. reports fixest's overall (not within-transformed) R^2, which nets out the region and country-year fixed effects and is mechanically much closer to 1, so the two are not directly comparable in level.

7. S5. Table VII: Aid, Oil, and Regional Favoritism

Table 6 reports the authors' aid/oil heterogeneity test (p. 1024), asking whether favoritism responds more to a leader's incentives when a country receives more foreign aid or earns more from oil, and whether that response depends on the strength of political institutions. Columns (1)-(2) interact the birth-region dummy with Aid_{ct-1} and Oil_{ct-1} separately; Columns (3)-(4) add a triple interaction with Polity_{ct-1}, testing whether the aid/oil channel itself is weaker where institutions are stronger. This table was already fully built by this project's own pipeline (script and fitted models) but had not yet been added to this document; it is included here for completeness against the authors' seven-table set.

This table diverges from the authors' own more than any other in this document. The authors' central finding, a positive and significant Leader × Aid interaction in Column (1) (0.008, p<0.01), does not reproduce: this replication's Column (1) estimate is smaller and not itself significant (0.003, SE 0.004). Column (3), which keeps the same interaction alongside the Polity triple interaction, shows the same pattern: the authors' Aid interaction there is significant (0.019, p<0.05), this replication's is not (0.008, SE 0.016). Column (2)'s Oil interaction diverges in the other direction: the authors' own estimate is small and not significant (0.000, SE 0.002), while this replication's is negative and significant (-0.008, $p<0.01$), the only term in this table where the replication is significant and the authors' own is not. The baseline Leader_{t-1} coefficient is not significant in either panel in Columns (1) and (3), but diverges more in Columns (2) and (4), where this replication's estimate turns significant (0.090, $p<0.01$, and 0.245, $p<0.05$) against the authors' own small, insignificant values. The two triple interactions, Aid $\times$ Polity and Oil $\times$ Polity, keep the authors' negative sign in three of the four relevant column entries but are not significant in this replication's version of either, where the authors report one of the two (Aid $\times$ Polity, Column 3) as significant. Given how much this table diverges, its result is best read as suggestive at most, not a robust confirmation of the aid/oil heterogeneity channel HR 2014 report.

Reading the columns economically. This table asks whether external resource windfalls, over which a national government has discretion but which did not have to be raised from its own citizens, are disproportionately captured for political rather than developmental use. Column (1)'s Leader $\times$ Aid interaction is the paper's aid- fungibility result: foreign aid is meant to be spent on the purposes donors attach to it, but if a marginal dollar of aid frees up an equal dollar of the government's own revenue for other uses, aid becomes fungible with general-purpose funds, and the authors' positive interaction says that freed-up margin disproportionately reaches the leader's own region rather than being spread nationally. Column (2)'s Oil interaction is the natural comparison case, oil rents being a domestic windfall rather than a donor transfer, and the authors' near-zero, insignificant estimate there is consistent with oil revenue being harder to redirect at the margin, whether because it is already embedded in existing fiscal and budgetary institutions in a way project aid is not, or because this project's own Oil measure is a coarser, country-level proxy than the region-specific extraction data a cleaner test would need. Columns (3)-(4) ask whether either channel is itself conditional on institutional quality: a negative Aid $\times$ Polity interaction says the aid-capture channel is weaker where political constraints are stronger, the same institutional-brake story Table V's Polity interaction tells for favoritism generally, extended here to say that stronger institutions do not just reduce favoritism on average, they specifically blunt a leader's ability to convert an aid windfall into regional patronage.

Table 6: Table VII replication, HR 2014's aid/oil heterogeneity specification, in the main paper's Orig./Repl. layout and the authors' own row order (p. 1024). Standard errors in parentheses, clustered by leader spell.

	(1)		(2)		(3)		(4)	
	Light$_{ict}$		Light$_{ict}$		Light$_{ict}$		Light$_{ict}$	
	Orig.	Repl.	Orig.	Repl.	Orig.	Repl.	Orig.	Repl.
Leader$_{t-1}$	-0.019	0.027	0.020	0.090***	0.086	0.102	0.118	0.245**
	(0.015)	(0.035)	(0.022)	(0.030)	(0.073)	(0.148)	(0.084)	(0.111)
Leader × Aid$_{ct-1}$	0.008***	0.003			0.019**	0.008		
	(0.002)	(0.004)			(0.009)	(0.016)		
Leader × Oil$_{ct-1}$			0.000	-0.008***			0.010	-0.014
			(0.002)	(0.003)			(0.008)	(0.011)
Leader × Polity$_{ct-1}$					-0.121	-0.121	-0.109	-0.208
					(0.074)	(0.165)	(0.094)	(0.129)
Leader × Aid$_{ct-1}$ × Polity$_{ct-1}$					-0.019*	-0.005		
					(0.010)	(0.018)		
Leader × Oil$_{ct-1}$ × Polity$_{ct-1}$							-0.014	0.009
							(0.010)	(0.013)
Number of regions	38,427	26,709	38,179	38,128	38,427	26,691	37,851	37,782
Observations	690,495	401,490	645,396	605,470	684,213	397,666	641,410	601,974
R^2	0.319	0.965	0.335	0.975	0.320	0.965	0.335	0.975
Region FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country-Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes. Fixed effect regressions using annual data for subnational regions between 1992 and 2009. Leader$_{ict-1}$ is a dummy variable equal to 1 if region i is the birth region of the political leader in $t-1$, and 0 otherwise. Polity$_{ct-1}$ is the Polity2 score, rescaled so that it ranges from 0 to 1, with higher values implying stronger political institutions. Aid$_{ct-1}$ is the log of net overseas development assistance per capita plus 1. Oil$_{ct-1}$ is the log of oil rents per capita plus 1. Standard errors are adjusted for leader clustering. ***, **, * indicate significance at the 1%, 5%, and 10% levels, respectively. (HR 2014, Table VII note, p. 1024, adapted.) This replication's own additions: Aid is built from QoG's OECD-DAC gross ODA series, IHS-transformed following Levy-Yeyati, Panizza and Stein (2007) as the authors specify; Oil is QoG's oil-rent series, log1p-transformed. Orig. R^2 is the authors' own reported R^2; Repl. reports fixest's overall (not within-transformed) R^2, which nets out the region and country-year fixed effects and is mechanically much closer to 1, so the two are not directly comparable in level.

Data and Code Availability

All data are drawn from open sources: subnational boundaries from GADM (GADM, 2018), nighttime lights from DMSP-OLS (Earth Observation Group, Payne Institute for Public Policy, Colorado School of Mines, 2013), leader identities and tenure from Archigos (Goemans et al., 2009), birthplaces from PLAD (Bomprezzi et al., 2024) supplemented by a Wikidata scrape (Wikidata, 2026), population from GPWv3 and GPWv4 (CIESIN, Columbia University and CIAT, 2005; CIESIN, Columbia University, 2018), and the Table V/VI/VII covariates from the Quality of Government dataset, the Penn World Table (Heston et al., 2012), the Polity IV project (Marshall et al., 2010), Barro and Lee's schooling series (Barro and Lee, 2013), Alesina et al.'s linguistic fractionalization (Alesina et al., 2003), Alesina and Giuliano's family-ties measure (Alesina and Giuliano, 2014), OECD development assistance statistics (OECD, 2013), and the World Bank's adjusted net savings series (World Bank, 2013). The complete pipeline that builds every table in this document, and in the main paper, is openly available in the replication repository (https://github.com/ofurkancoban/ofurkancoban-Regional_Favoritism_Replication). Table I, III, V, VI, and VII replicate five of the authors' seven main-text tables that the main paper's page limit leaves no room for; they are reported here in full rather than in the paper, and none of them changes that paper's conclusions.

Declaration of AI Use

During the preparation of this work, the author used Anthropic Claude Sonnet 5 for coding assistance in building the replication pipeline and for English-language improvement of the manuscript text. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of this publication.

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